Identifying Topological Hubs and Functional Modules in the *Tulipa uniflora Chloroplast Genome*

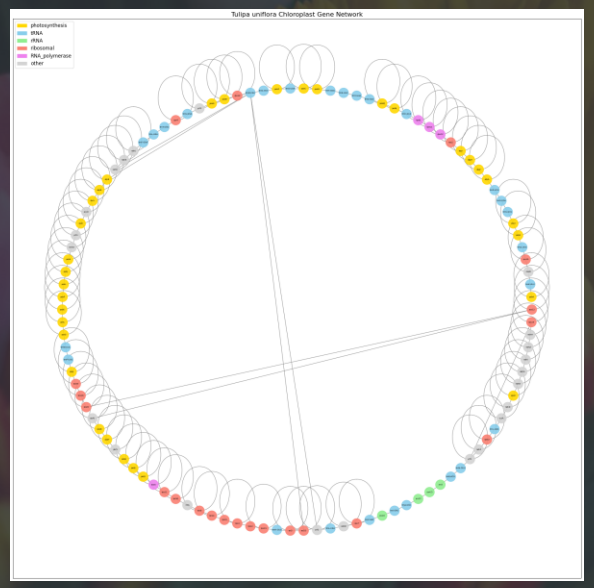
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ABSTRACT / INTRODUCTION

This project models the 152,254 bp chloroplast genome of Tulipa uniflora as a gene adjacency network. Genes are treated as nodes and physical neighbors along the circular plastome are linked by edges.

A network-science view makes it possible to identify structural hubs, visualize functional clustering, and ask whether the plastome is organized randomly or in a modular way.

FIGURE 1 | GENOMIC ADJACENCY NETWORK



110 genes and 188 adjacency edges are arranged in a circular layout that mirrors the plastome. Distinct color blocks show that genes with shared functions occupy localized regions rather than appearing uniformly around the genome.

SCIENTIFIC HYPOTHESIS

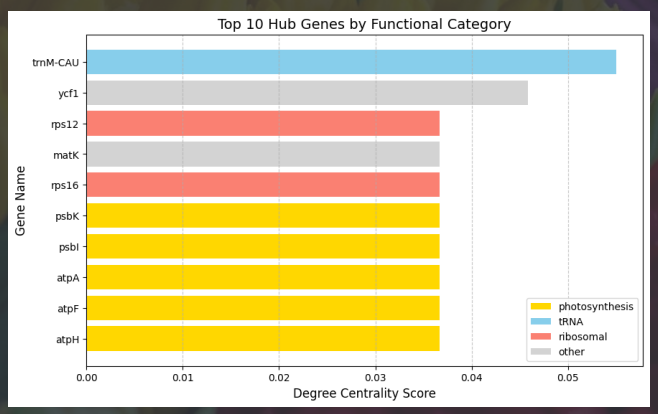
Genes from the same functional class—especially photosynthesis genes—should cluster locally and show higher internal edge density than more housekeeping-oriented groups such as tRNAs.

Prediction: the photosynthesis subgraph will be denser than the tRNA subgraph after normalizing by group size.

METHODOLOGY

- Data source: complete chloroplast genome of Tulipa uniflora (NCBI accession PP861170.1).
- Graph construction: 110 unique genes as nodes and physical adjacency links as undirected edges.
- Network metrics: degree centrality for hub detection plus subgraph analysis for modularity.
- Functional categories: photosynthesis, tRNA, rRNA, ribosomal, RNA polymerase, and other.
- Tools: Python, Biopython, NetworkX, Matplotlib, and pandas.

FIGURE 2 | TOP 10 HUB GENES



trnM-CAU and ycf1 are the most central genes, followed by ribosomal and photosynthesis-related hubs such as rps12, matK, psbK/psbI, and atpA/F/H.

Photosynthesis: 1.56 edges/gene (50/32)
tRNA: 0.40 edges/gene (11/27)
Takeaway: the photosynthesis module is nearly 4× denser, supporting non-random clustering.

RESULTS / INTERPRETATION

- Figure 1 reveals visible "color blocks," meaning genes with related functions are physically clustered around the circular plastome.
- The strongest hubs are trnM-CAU (initiator tRNA) and ycf1, which likely act as key structural anchors or bridges between genomic regions.
- Many high-centrality genes also support translation, ribosome assembly, or photosynthetic energy conversion—functions that benefit from coordinated local organization.

WHY IT MATTERS

A network view of the plastome makes spatial organization measurable. Dense neighborhoods of photosynthesis genes suggest selective pressure for co-regulation and efficient metabolic coordination, while the broader dispersion of tRNAs supports genome-wide translation needs.

Together, these patterns indicate that genomic proximity in the chloroplast is functionally meaningful rather than incidental.

CONCLUSION

- The Tulipa uniflora plastome is non-randomly organized.
- Photosynthesis genes form a robust functional module with high internal connectivity.
- tRNA genes are more dispersed, providing translation coverage across the genome.
- Network analysis highlights a biologically interpretable map of chloroplast architecture.

REFERENCES

1. Kim S. et al. (2024). The complete chloroplast genome of Tulipa uniflora (Liliaceae) in Mongolia.
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